

The Detector, Understood: Single-Photon Phenomena in the Rope Framework, in Plain Language

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Companion to the measurement-sector campaigns, FND-STRAND-009 through 029 (releases v3.7.0 onward), including the third grant. Written for a reader outside the project. Canonical paper: [papers/the_detector_understood.pdf](#); this markdown is its living source. Every claim referenced here has a registered entry in `claims.yaml`, an executable benchmark, and pre-committed pass criteria in `analysis/`.

The standard evidence for quantized light: a six-item scoreboard

Physics textbooks support quantized light with a standard list of experiments. Since the framework adopted its third assumption (the quantum arrives whole, explained below), each item on that list now has a definite, registered status. Here is the scoreboard, in plain terms.

1. Beam-splitter correlations (Grangier-Roger): DERIVED. Send one photon at a half-silvered mirror and exactly one side clicks, 50/50, never both. The classical machinery measurably could not do this (the split packet clicked neither side). With the third assumption plus the framework's existing probability rule, it becomes one line of arithmetic: the whole quantum lands on one side, with odds given by the energy each side receives.

2. Antibunching: DERIVED, with a bonus theorem. A true single-photon source never produces two clicks at once. The framework proves something stronger and practical: the antibunching measurement is immune to loss. No amount of detector inefficiency or imbalance can fake or destroy it, which is exactly why the historic experiments could trust their result. The famous measured value of 0.18 gets a sharper reading too: it is not "imperfect antibunching" but a direct measurement of how often the source accidentally emits two photons.

3. Photon-count statistics: DERIVED, both regimes. Laser light clicks randomly (Poisson); lamp light clicks in bunches (Bose-Einstein). Both fall out of the framework's own detector, and a small theorem explains a piece of history: for ordinary light, the smooth-wave description and the counting-quanta description give IDENTICAL statistics. That is why classical optics thrived for a century without photons, and why only special quantum light sources could ever force the issue.

4. Delayed choice: DERIVED AS A NON-PARADOX. Wheeler's famous puzzle – decide how to measure AFTER the photon is in flight, and it seems to retroactively "choose" wave or particle behavior – dissolves because the framework never had a particle that chose anything. The wave

takes both paths in every configuration; the only event is the click at the end; a late change to the apparatus just changes what the still-flying wave will meet. As a bonus, the celebrated wave-particle trade-off formula ($V^2 + D^2 = 1$) comes out as four lines of energy bookkeeping.

5. Hong-Ou-Mandel interference: PINNED, not yet explained. Two identical photons meeting at a splitter always exit together – the deepest two-photon effect below Bell tests. The framework derives its own exact shortfall: its wave picture can reach at most half the measured effect, its granular picture predicts none of it, and the missing ingredient is located precisely (a correlation in how two identical quanta are DELIVERED, not in how they travel or are detected). The shape of a fix is specified, with its own kill conditions pre-drafted, and deliberately not yet adopted.

6. Heralded single photons, and the quantum eraser: HONESTLY DEFERRED. Heralding (using one photon of a pair to announce its twin) mostly needs something the framework has not built yet: a mechanical model of the SOURCE, not the detector. And the entangled quantum eraser sits behind the framework's oldest proven boundary: a counting model cannot produce Bell-type correlations, full stop, and the registry says so rather than promising otherwise.

The pattern across the scoreboard: four items converted from quantum mystique into detector bookkeeping, one bounded with its missing piece located to the millimeter, one deferred with the reason named. Every verdict traces to a registered claim with an executable check.

Three questions this framework can now answer

Before the detail, here are the three results in their simplest form. Everything below unpacks them; every sentence traces to a registered, benchmarked claim.

1. How does a single-photon detector actually work? A click is a real mechanical event: a tiny knot forming on a strand when enough energy gathers at one spot. The detector is a threshold device, like a mousetrap. The surrounding medium (the weave) is a warm bath that constantly jiggles the strand, and the rate of clicks follows the classic escape-over-a-barrier law. The remarkable part: the “attempt rate” in that law is not a fitted number. It is the bath's own minimum vibration frequency, the strand mass scale. The medium itself is the detector's metronome, and this held up under two independent statistical methods with opposite failure modes.

2. Why does a small detector seem to have memory, when nothing in it remembers anything? Freshly reset small detectors click fast at first and then settle down, which looks like aging or memory. We hunted for the memory with purpose-built instruments and found nothing changing: temperature flat, forces flat, no starting condition predicting anything. The answer came from changing one line: start the detector in true equilibrium with its bath instead of switched on cold, and the effect vanishes completely. The “memory” was a switch-on transient, the brief extra jostling any system feels while it and its bath get acquainted. Nothing inside ever remembers; the appearance of memory is the echo of the power-up. And detector size hides the effect by pure arithmetic: a large detector is many independent click sites racing each other, the first click always comes early, and early clicks only ever sample the settled behavior. We proved the exact law: the large detector's statistics are the small one's raised to the size ratio, with zero adjustable constants.

3. How can single photons, fired one at a time over hours, build up an interference pattern, as if each one interferes with itself? Because nothing pointlike ever travels. Each emission is a wave that goes through both slits every single time and lays the full striped energy pattern on the screen. The dot is not the photon arriving; the dot is the detector firing, a threshold event that is pointlike because that is what tripping a threshold is, like one ping of rain on a tin roof. The odds of the dot landing at any spot are proportional to the wave energy delivered there, so each photon's single dot is a sample drawn from the same standing striped pattern. Fire ten thousand photons over ten hours and the dots trace the stripes, because every dot was drawn from the same map. Nothing needs to remember anything between shots, and no photon ever "chooses a slit," because no photon ever travels as a particle. The wave delivers the odds; the bath delivers the dot.

One honest sentence to complete the picture: this account is established for the plain double-slit, and the framework states exactly where it ends. Splitting one photon's energy between two separated detectors is beyond the classical machinery (the funneling step, described near the end of this document), and that boundary is measured and registered, not glossed.

The question in one sentence

When a detector clicks in the dark, with no signal present, what sets the timing of those false clicks, and does the pattern carry any fingerprint of the physics underneath?

In this framework a detector click is a physical event: a small knot nucleates on a strand when enough energy gathers in one place. The "darkness" around the detector is not empty. It is the weave, a background of strand vibrations that acts as a heat bath. So the dark-count question becomes a concrete mechanical one: how does a warm weave kick a strand over its threshold, and how often?

The short answer

A detector's dark-count behavior has exactly two regimes, and we can now state both from first principles, with every constant accounted for.

Steady state. Leave any detector running, large or small, and its clicks are textbook random: a steady Poisson stream, the same odds at every moment. The rate follows the classic escape-over-a-barrier law, and the "attempt frequency" in that law is not a fitted number. It is the weave's own minimum vibration frequency, the band gap, which is the strand mass scale. The metronome of the detector is the medium itself. This identification survived two independent statistical methods with opposite failure modes, which is about as hard a test as a simulation result can be given.

Just after switch-on. Power up, reset, or suddenly isolate a small detector, and for a limited time its clicks are NOT steady. The click rate starts elevated and relaxes downward to the steady value over one characteristic epoch, the time it takes the detector and its bath to become properly acquainted. During this window the waiting times between clicks are measurably non-exponential. After the window closes, the detector is ordinary forever.

And size matters in a precise way. The switch-on fingerprint is strongest in the smallest detectors and vanishes in large ones. Not because large detectors settle faster. They do not; the settling time is a property of the medium and is the same at every size. Large detectors hide the fingerprint by arithmetic: a big detector is many independent nucleation sites racing each other, the first click always arrives early, and early clicks sample only the flat beginning of each site's curve. We proved the exact law for this. Take the measured survival curve of the small reference detector and raise it to a power equal to the size ratio. That is the whole formula. Nothing in it is fitted; the one quantity you would expect to tune, the number of independent sites, cancels out of the algebra. The formula predicted three independently measured sizes inside their error bars with zero adjustable constants, and it now carries the corpus's Derived status, the highest tier.

How we got it wrong before we got it right

The honest history matters, because the wrong turns are load-bearing.

For seven sessions the falling click rate looked like a mystery. It behaved like memory, so we hunted for the memory with purpose-built instruments, each with its pass and fail conditions locked before the data existed. A thermometer on the bath: temperature flat to five parts in ten thousand. A meter on the force the bath delivers: flat to six parts in ten thousand. A census of 256 starting configurations: no measurable property of the start predicted a detector's fate. A complete deterministic theory of the effect: dead at its own first checkpoint, kept on display in the registry as required. At that point the phenomenon had been replicated three times on independent data and was invisible to every natural variable.

The resolution came in two steps. First, the size study and the power-law derivation showed the per-site memory is real, permanent, and intensive, and that large systems simply never look at it. Second, and decisively, one final experiment changed a single line: instead of starting each simulation with the strand perfectly cold and uncorrelated with its bath (the "product state," which is what the code had always done), we prepared the true joint equilibrium of strand and bath together. The falling rate vanished completely. Flat clicks, exponential waiting times, at exactly the rate the old ensembles had always been decaying toward.

The mystery had been hiding in the first line of every script. The strand started at zero. Physics calls the consequence "initial slip": a system switched on uncorrelated with its bath feels a brief excess of effective noise while correlations build, even though every stationary quantity (temperature, force intensity) reads constant the whole time. Our instruments were not wrong. They were measuring exactly the things that do not change. The transient lives in the correlations those instruments integrate out.

So the earlier confusion was necessary. Each flat instrument was an exclusion that pinned the effect into the one corner where it actually lives, and the dead theory exposed the internal mixing of the bath that is precisely the mechanism by which the slip relaxes.

What an experimentalist could do with this

Prediction 11 of the framework now reads as a two-part statement that a lab could in principle test on small single-photon detectors or similar threshold devices:

1. **The quench fingerprint.** After a reset or power-up, a sufficiently small, well-isolated detector should show dark counts whose waiting times are non-exponential for one bath-correlation epoch, with the apparent rate drifting down while the device's measured temperature holds constant. A rate drift at constant temperature is the distinctive part; ordinary thermal-drift explanations cannot produce it.
2. **The size law.** Compare detectors of different sizes prepared the same way. The deviation from Poisson should shrink with size following the exact power law: the large detector's survival curve is the small one's raised to the size ratio. No fitting allowed; the curve for the small device fully determines the prediction for the large one.

And the boring regime is a prediction too: in steady operation, every size should be clean Poisson, with an attempt-rate prefactor set by one structural scale of the medium, the same scale that sets the decoherence floor in Prediction 10. Two independent instrument-facing predictions pinned to a single number is the kind of rigidity that makes a framework falsifiable rather than flexible.

The scale caveat, stated plainly

All of this is a statement about the model's own units. The framework does not yet fix the absolute physical scale of the strand mass, so these are predictions about the SHAPE and the RELATIONSHIPS of detector statistics (two regimes, a size power law, drift at constant temperature, one shared prefactor scale), not yet about a frequency in hertz. The shape claims are falsifiable now; the absolute calibration is registered as open.

The sequel: one click, and the edge of the story

After the two-regime picture was settled, we chased the last loose thread: why does one photon's worth of energy produce exactly one click, never two? Three sessions later the answer arrived, and it came in two parts.

Part one: the click is a cliff. When we finally modeled the photon correctly, as a concentrated packet rather than a faint glow spread over the whole screen (two earlier designs failed for exactly that reason, and the failures are kept on the books), the detector's response turned out to be a sharp threshold: a full packet clicks about half the time, and a packet with HALF the energy clicks never. Not rarely. Never, in every trial. That settles the one-click question almost embarrassingly simply: split one photon's energy between two places and neither place can click, so a double click was never possible. No hidden bookkeeping needed. The staircase is so steep that half a step is the same as no step.

Part two: the honest edge. But that simplicity exposes something the framework must own. In real laboratories, a single photon sent through a beamsplitter DOES click, on exactly one side, about half the time each. Our classical machinery, asked the same question, says neither side clicks. So the framework's detector story, for all its success, cannot yet do the one thing every beamsplitter experiment shows: gather a split photon's full energy back into one spot. We call that missing move the funneling step, and rather than paper over it, it is registered as a measured boundary, with the experiments that live beyond it (which-path tests, delayed choice, the quantum eraser) explicitly marked as not yet in reach. The leading idea for crossing the boundary is a single minimal assumption, that a quantum's energy is always delivered whole or

not at all, which together with the framework's existing probability rule would reproduce the laboratory statistics. Whether to adopt that assumption is a decision recorded as pending, not smuggled in.

A theory that can say “here is exactly where my story ends, and here is the measurement that proves it” is doing something most theories never manage. The edge is not a defeat. It is a surveyed border, and surveyed borders are where the next expedition starts.

Where to look in the corpus

- The campaign: claims FND-STRAND-009 through FND-STRAND-021 in `claims.yaml`, each with its benchmark under `benchmarks/foundations/`.
- The cornerstone law (Derived): FND-STRAND-020, `benchmarks/foundations/strand_poissonization.py`.
- The switch-on resolution: FND-STRAND-021, `benchmarks/foundations/strand_switch_on.py`.
- Pre-committed pass criteria for every session: `analysis/STRAND0xx*_bars_LOCKED.md`.
- The kills and the buried theory, kept on display: FND-STRAND-015, 016, and 017.
- The exclusivity arc and the surveyed edge: FND-STRAND-022, 023, and 024, with the limit statement now in `papers/falsifiable_predictions.pdf` (“A Registered Limit — The Funneling Step”).
- The third grant and the scoreboard's derivations: FND-STRAND-025 (the grant and the beamsplitter), 026 (antibunching and the faithfulness theorem), 027 (Mandel counting and the correspondence theorem), 028 (delayed choice and the duality relation), 029 (the Hong-Ou-Mandel pin).

Every number quoted above can be recomputed from the archived datasets by running the corresponding benchmark.